Theme 1: FUNCTIONS

Note Title 2/5/2018

Unit 1.1 Numbers, Intervals and Inequalities.

- In this lecture we focus on Inequalities.
- Read Stewart pp. A2–A6
- How you write Mathematics is important, i.e., Notation and symbols are important.

- $\Longrightarrow$ Know how to use the imply sign
- Intervals and set builder notation

$\cap$ Intersection of 2 sets (AND)

$\cup$ Union of 2 sets (OR)

- To solve an inequality: use a graph where possible.
- To solve a quadratic inequality: ALWAYS use a graph.
- p. A4: Know the Table of Intervals and the Rules of Inequalities.

Rational numbers: (p. A2)

Examples:

1) Write in interval notation

a) $x>3$ and $x<5$

b) $x<5$ or $x>3$

c) $x<3$ or $x>5$

d) $x<3$ and $x>5$.

SOLVE FOR x:

2a) $x(x+1)=0$

2b) $x^2-1=0$

Inequaties: Rules (p. A4)

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Examples:

(1) $x^2 < 2x+8$

(2) $x^3+3x < 4x^2$

(3) $\frac{1}{x} < 4.$

(4) $\frac{x+1}{2-x} \geq 0$

(5) $-x^2-x-1>0$